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Initial results of a long-term coral reef monitoring program: Impact of Hurricane Hugo at Buck Island Reef National Monument, St. Croix, U.S. Virgin Islands

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Abstract: Fixed position linear transects were established in early 1989 at Buck Island Reef National Monument, St. Croix, U.S. Virgin Islands. On September 17–18 of that year Hurricane Hugo, the most severe cyclone to impact the area in over 60 years, passed directly over the island bringing hurricane-force winds for over 12 h and maximum estimated wind speeds of $70 \text{ m} \cdot \text{s}^{-1}$ (160 mph). Despite the severity of the storm, damage to coral reefs was extremely localised and concentrated mainly on reefs open to its southeasterly direction of approach. The southeast reef front at Buck Island was razed to substrate level between the surface and 7 m depth and the reef crest behind it was smothered in a 1 m deep berm of broken coral rubble. At a site on the north backreef, however, only a slight loss of coral cover was detected which was more than compensated for during 1990–91. At a south reef site which was in 8–10 m depth, just outside the region of severe damage, coral cover was reduced by 40–46% on three out of four transects. Shifts in community structure were detected by multi-dimensional scaling of Bray-Curtis similarity measures and by *k*-dominance curves, but not by the Shannon diversity statistic (H'). Coral cover had returned to approximate pre-hurricane levels by June 1991, but community composition remained distinct. One of the four transects at this south site was apparently not significantly damaged during the hurricane. Such spatial variability may affect recovery rates, since pockets of relatively undisturbed benthos may provide seed populations for recruitment into adjacent, more severely damaged areas. Hurricane Hugo did not appear to cause an immediate increase in species diversity by differential mortality of the dominant species in the community. This result is consistent with previous studies of the impact of less severe storms on St. Croix, but contrary to several reports of hurricane impact in other areas.

Key words: Coral reef; Community structure; Hurricane; Disturbance

INTRODUCTION

Over the past two decades the paradigm that coral reefs represent stable, climax communities has been questioned by proponents of non-equilibrium models of organisation (Connell, 1978; reviewed by Brown & Howard, 1985). Hughes & Jackson (1985) showed that coral population turnover may be relatively rapid, even when community

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structure remains apparently stable. These findings will influence predictions of the effects of natural and man-induced perturbations on reefs and their subsequent recovery (Porter et al., 1982; Dustan & Halas, 1987). The extent of damage from hurricane events and processes of recovery have been shown to be highly variable between coral reef sites (Porter et al., 1981; Rogers et al., 1982; Knowlton et al., 1990). Also, the frequency of hurricane impacts is low enough that we are on only just beginning to understand that hurricanes may play a key role in shaping coral reef community structure (Stoddart, 1974; Highsmith et al., 1980; Woodley et al.; Graus et al., 1984; Hughes, 1989).

Both local and regional disturbances on coral reefs are becoming more widely reported (Rogers, 1985; Brown et al., 1990). Although we are presently unable to state for certain that there have been any biological or ecological effects of global climate change (D'Elia et al., 1991), the need for comprehensive long-term monitoring of what may be particularly sensitive ecosystems (Jokiel & Coles, 1990) is now clearly recognized (D'Elia et al., 1991).

In 1988 the U.S. National Park Service initiated a long-term ecological monitoring program at coral reef sites in the Caribbean and Florida (Rogers, 1989). In 1989 Hurricane Hugo impacted two of the sites, making a direct hit on the island of St. Croix, U.S. Virgin Islands (Hubbard et al., 1991) and causing severe damage on nearby St. John (Edmunds & Witman, 1991; Rogers et al., 1992). This paper addresses the effects of the hurricane on the reefs of Buck Island, a protected National Monument off the north east coast of St. Croix ($17^{\circ} 47' \text{ N}$, $64^{\circ} 37' \text{ W}$). Damage from the hurricane, the most severe to impact the site since Hurricane San Felipe in 1928 (local eyewitness accounts), was concentrated on the forereef on the south and east sides of the island down to a depth of approximately 7 m. Elsewhere, damage was remarkably light, although fixed position linear transects on the north reef did show a significant change when assessed using the chain transect technique (Porter, 1972; Rogers et al., 1983), even though the percent surface area loss of live coral was only approximately 2%.

METHODS

Four fixed position 20-m long transects were established at each of two sites within the Buck Island bank-barrier reef system, described in detail by Bythell et al. (1989) and Gladfelter et al. (1991). Briefly, they can be characterized as follows:

Site 1. Located between 8 and 10 m depth on the forereef on the south reef of the island. A relatively diverse coral community, with moderately high structural relief produced mainly by live colonies of the boulder star coral *Montastraea annularis* (Ellis & Solander) and finger coral *Porites porites* (Pallas). Moderate coral cover (13–29% determined by chain transects).

Site 2. A protected, shallow (3 m) site on the backreef of the north side of the island. A less diverse coral community with very low structural relief; a fairly flat coralline

pavement with abundant medium to large sized colonies of the brain corals: encrusting *Diploria clivosa* (Ellis & Solander) and hemispherical *D. strigosa* (Dana). Relatively high coral cover (32–40% determined by chain transects).

The starting point of each transect within a site was haphazardly placed and transect lines were then run approximately parallel to the reef front. Coral cover and diversity along the transects was assessed using the chain transect technique (Porter, 1972; Rogers et al., 1983), which employs a three-dimensional approach to assessing relative percent cover of benthic organisms. This method was chosen because cryptic species and vertical or overhanging surfaces are assessed, substrates which will be omitted when using most benthic assessment techniques which involve planar, two-dimensional cover estimates.

Transects were marked with steel stakes (concrete reinforcing bar) placed at each end and in the center of the transect (0, 10 and 20 m), which were left in place between repeat surveys. During each survey, a fibreglass measuring tape was stretched taut between the markers to establish the line of the transect and horizontal distances along it. A chain with 1.3 cm links was then carefully placed over the surface contour immediately below the tape. For each linear metre, the number of chain links crossing all biota and substrate types were counted, and the substrate type recorded, including scleractinian corals and hydrocorals to species.

Transects within sites were treated as independent replicates, and changes over time assessed using repeated measures ANOVA. Changes are representative only of the fixed study sites and may or may not reflect general changes in reef community structure in adjacent areas. In order to assess the variability of the technique itself, each transect was repeated within a few days of each other in the Spring of 1989 (Gladfelter et al., 1991). While no significant differences were detected in total coral cover (paired *t*-test; $t = 0.73$, $p > 0.4$), rarer species were not consistently included between replicates. All such replicates were included in the repeated measures ANOVA to assess time-dependent changes.

Corals form clonal, variable-sized colonies and each chain-link of the transect covering a given species was therefore treated as an “individual” for the purposes of assessing coral community diversity statistics, rather than number of colonies encountered. Univariate measures (e.g. percent live cover, Shannon diversity H') have been shown to be less sensitive than multi-dimensional scaling (Kruskal & Wish, 1978) and certain graphical descriptors (*k*-dominance curves: Lamshead et al., 1983) in detecting changes in coral community structure over time (Warwick et al., 1990). These multivariate methods were therefore employed to examine the impact from Hurricane Hugo and subsequent recovery over the following 2-year period.

k-Dominance curves may be used to show differences in species diversity of two or more samples and is independent of any bias towards species richness or evenness, a problem which affects combined indices such as H' . Non-intersecting *k*-dominance curves indicate a difference in species diversity of two samples, with the upper curve

representing a less diverse sample. Multi-dimensional scaling ordination (MDS) was carried out on a matrix of Bray-Curtis similarity indices, derived from square-root transformed species cover data (Warwick et al., 1990).

RESULTS

The shallow forereef on the south side of the island was cleared to bare pavement in many areas, with severe destruction along the entire south and east reef front. Small coral fragments (1–5 cm), mostly *Porites porites* and the fire coral *Millepora complanata* Lamarck, resulting from this destruction were deposited in a rampart 1 m deep and 10–30 m wide which smothered much of the previous reef crest/backreef community. At the end of this study, 2 years after the hurricane, the rubble rampart was still intact and forms a physical reef crest 30 m landward of the pre-hurricane crest. Larger coral boulders and dead elkhorn coral *Acropora palmata* (Lamarck) stands were also flung up onto the rampart and into the backreef, but were more generally deposited down-slope to the base of the steep, shallow forereef at 7–8 m depth. Although the fixed transects in place on the south reef were in 8–10 m depth, outside the region of severe destruction, reductions of 9–14% coral cover occurred on three out of four transects, representing between approximately 40% and 46% of the pre-hurricane coral cover (Fig. 1). This did not result in a significant change over time however (repeated measures ANOVA; $F = 2.2$, $p > 0.1$), unless transect 3 was omitted ($F = 8.4$, $0.05 > p > 0.01$). Reduced coral cover was generally associated with a drop in diversity (Shannon H'), although no significant time-related changes could be determined ($F = 3.3$, $0.1 > p > 0.05$). All transects had low species diversity, reflecting the dominance in this area by *Montastraea annularis* and *Porites porites* (Table IA). No significant differences could be detected between the immediate pre- and post-hurricane percent cover of these dominant species (paired t -test; $t = 0.54$, $p > 0.5$ and $t = 1.46$, $p > 0.2$, respectively). Percent cover of the more fragile finger coral *P. porites* dropped substantially on transects 1, 2 and 4, however. Transect 4 showed consistently higher species diversity than the other transects, coupled with the lowest percent coral cover.

At site 2 (backreef, north Buck Island), a significant change in coral cover over time was detected ($F = 19.0$, $p < 0.01$), but not in Shannon H' ($F = 1.0$, $p > 0.4$). Coral cover appeared to drop slightly following the hurricane, but had risen above pre-hurricane levels on most transects by July 1991 (Fig. 2). As at site 1, coral diversity was generally low, reflecting dominance by another pair of species, *Diploria clivosa* and *D. strigosa* (Table IB). No significant differences were apparent between immediate pre- and post-hurricane percent cover of these species (paired t -test; $t = 0.25$, $p > 0.8$ and $t = 1.48$, $p > 0.2$ for *D. clivosa* and *D. strigosa*, respectively).

From k -dominance curves for the two sites (Fig. 3), it appeared that at site 1 species diversity decreased slightly following Hurricane Hugo and had not returned to pre-hurricane levels by July 1991. The curves appear to intersect for all surveys car-

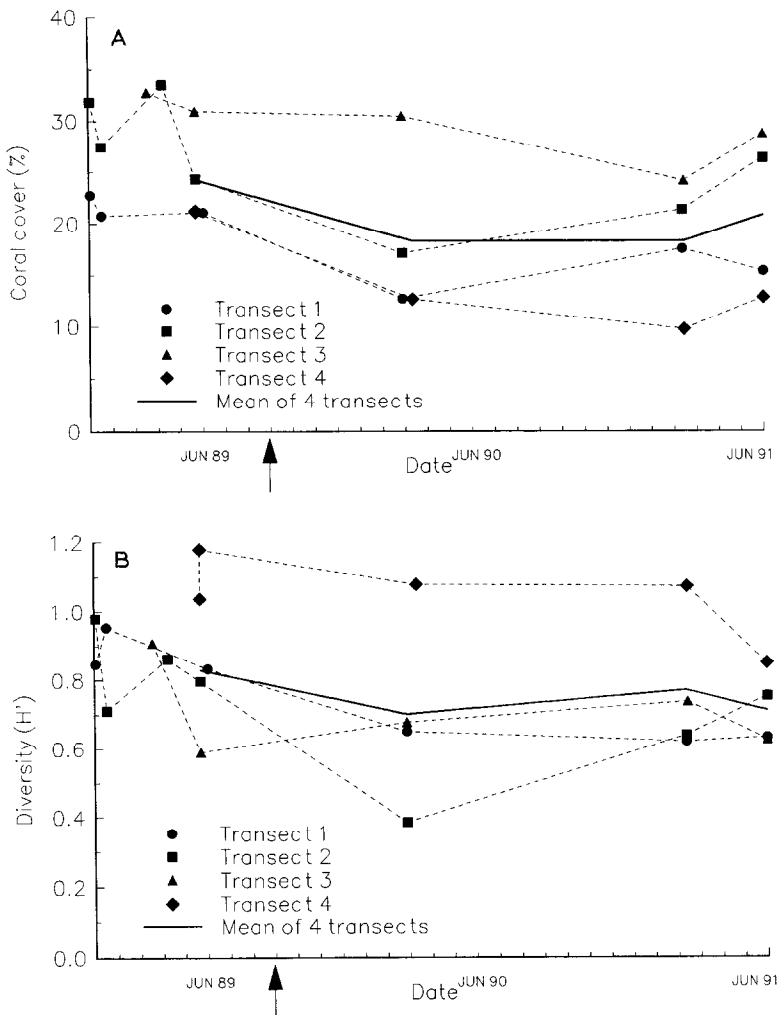


Fig. 1. Percent coral cover (A) and Shannon diversity index H' (B) over time for each of the four replicate transects at site 1. Arrow shows the time of impact of Hurricane Hugo. Solid lines show mean values where all four transects were surveyed at the same time of year.

ried out at site 2, indicating that no unambiguous distinction can be made between pre- and post-hurricane diversity.

At site 1, MDS ordinations of Bray-Curtis similarity measures for coral cover showed a clear shift in coral composition on transects 1, 2 and 4 following the hurricane (Fig. 4A). That is, pre-hurricane survey samples cluster independently of those carried out post-hurricane. Interestingly, the direction of shift varies, indicating that the effects of the hurricane were not uniform, but perhaps depended on the original species composition of the transect. The lack of a clear shift on transect 3 supports the find-

TABLE I

Coral species percent cover data for pre-hurricane (1989) and post-hurricane (1990, 1991) surveys. Only data from surveys carried out in spring of each year are included; repeat surveys were carried out within a few days of each other by different observers in 1989 (89a and b), ND, no data.

Species	Percent cover															
	Transect 1				Transect 2				Transect 3				Transect 4			
	89a	89b	90	91	89a	89b	90	91	89a	89b	90	91	89a	89b	90	91
A. Site 1																
<i>Acropora cervicornis</i>	0.3	0.3	-	-	0.2	0.1	-	-	-	-	-	-	0.7	0.9	0.1	0.1
<i>Agaricia agaricites</i>	0.1	-	0.6	0.7	0.1	0.8	-	-	-	-	-	-	0.1	1.0	0.7	-
<i>Colpophyllia natans</i>	-	-	-	-	3.2	1.8	0.2	0.4	-	-	0.5	-	-	-	-	-
<i>Eusmilia fastigiata</i>	0.4	0.5	-	-	-	-	-	-	-	-	-	-	-	-	-	-
<i>Millepora alcicornis</i>	0.1	0.2	0.4	-	-	-	-	-	0.6	0.3	-	-	0.1	-	-	-
<i>M. complanata</i>	-	0.4	-	-	-	-	-	-	-	-	-	-	-	-	-	-
<i>M. squarrosa</i>	-	-	-	-	-	0.1	-	-	-	-	-	-	-	-	-	-
<i>Montastraea annularis</i>	14.3	12.0	10.4	14.0	18.0	15.6	15.3	16.5	6.1	6.8	8.9	7.7	5.0	5.8	7.3	5.0
<i>Porites astreoides</i>	-	-	-	-	-	-	-	0.2	-	-	-	-	-	-	-	0.2
<i>P. furcata</i>	-	-	-	0.1	0.1	0.4	-	-	-	0.8	-	-	1.5	1.8	1.2	0.9
<i>P. porites</i>	7.5	7.4	1.3	2.8	10.3	8.6	1.6	4.2	24.2	16.4	21.1	15.8	13.2	11.5	3.3	3.4
Total coral cover	22.7	20.7	12.7	17.5	31.9	27.5	17.1	21.3	31.0	24.3	30.5	24.1	21.2	21.1	12.6	9.7
B. Site 2																
<i>Agaricia agaricites</i>	-	ND	-	-	-	-	-	0.1	-	-	-	-	-	0.2	-	-
<i>Diploria clivosa</i>	17.1	ND	16.7	19.2	15.2	15.9	13.3	14.3	27.9	29.0	25.6	28.8	27.9	27.3	25.6	27.6
<i>D. strigosa</i>	14.8	ND	13.0	13.5	7.6	8.6	9.8	12.1	6.3	7.9	6.5	5.7	8.1	8.6	6.5	9.9
<i>Isophyllastrea rigida</i>	-	ND	-	-	-	-	-	-	0.2	0.3	-	-	-	-	-	-
<i>Porites astreoides</i>	-	ND	0.9	-	-	0.1	-	-	-	0.3	1.4	0.6	0.6	0.9	1.4	0.1
<i>Siderastrea radians</i>	-	ND	-	-	-	-	-	-	-	-	0.3	-	0.1	-	0.3	-
Total coral cover	31.9	ND	30.6	32.7	22.8	24.5	23.2	26.5	34.4	37.5	33.9	35.0	36.8	37.0	33.9	37.6

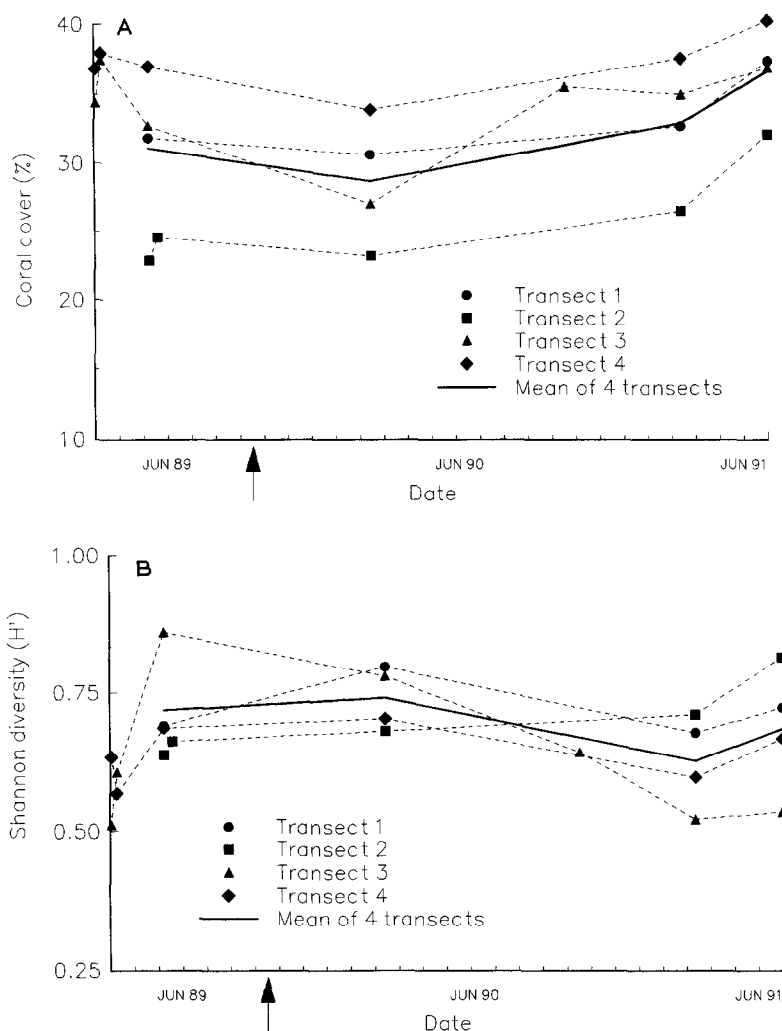


Fig. 2. Percent coral cover (A) and Shannon diversity index H' (B) over time for each of the four replicate transects at site 2.

ings from coral cover and Shannon diversity data, indicating that no significant changes in community structure occurred on this particular transect due to the hurricane. No consistent shifts over time could be detected in MDS ordinations of coral cover data at site 2 (Fig. 4B), a result that contrasts with the significant changes seen in percent coral cover. This suggests that the post-hurricane increases in coral cover (Fig. 2) were a result either of growth of colonies previously present, or (perhaps less likely) the recruitment of individuals of similar species composition to that already present on each transect.

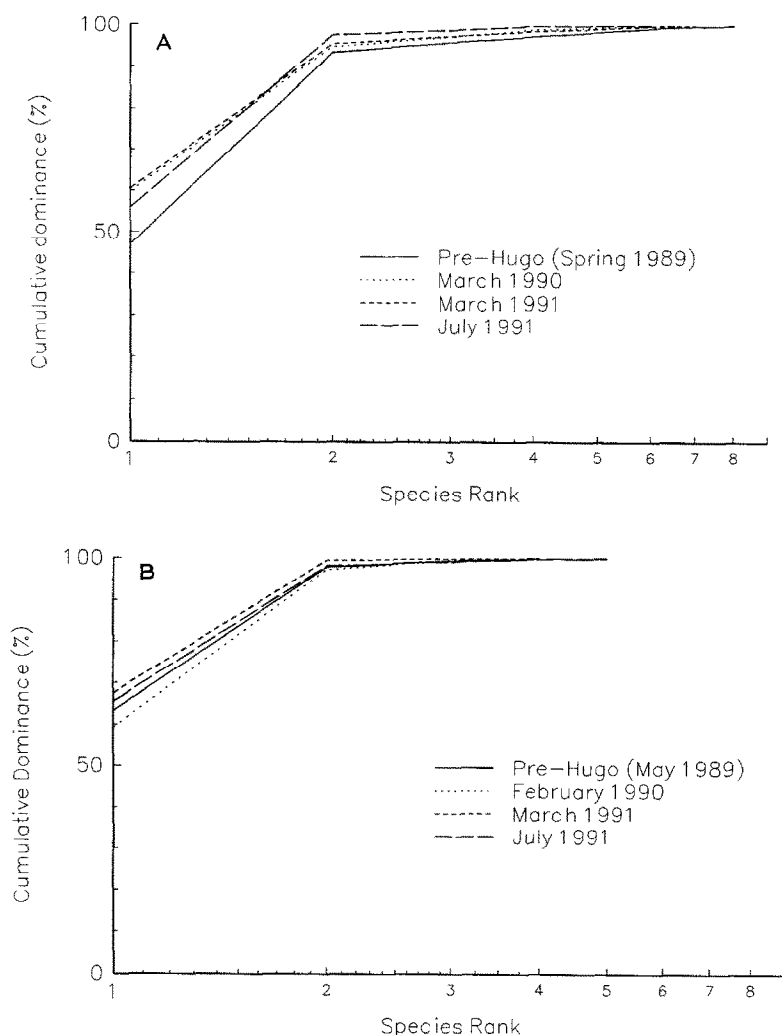


Fig. 3. k -dominance curves generated from surveys carried out at different times at site 1 (A) and site 2 (B). Cumulative percent dominance of each species is plotted against its rank, on a log scale.

DISCUSSION

This study highlights the spatial variability in hurricane impact, which was seen at two scales: there was a dramatic localization of effects between specific reef zones (Gladfelter et al., 1991), and at one site we were also able to detect variability on the scale of one to a few metres. The most seriously impacted reef area was the south reef crest and forereef down to a depth of approximately 7 m (Hubbard et al., 1991; Gladfelter et al., 1991). Hurricane damage on the north of the island was far less severe and

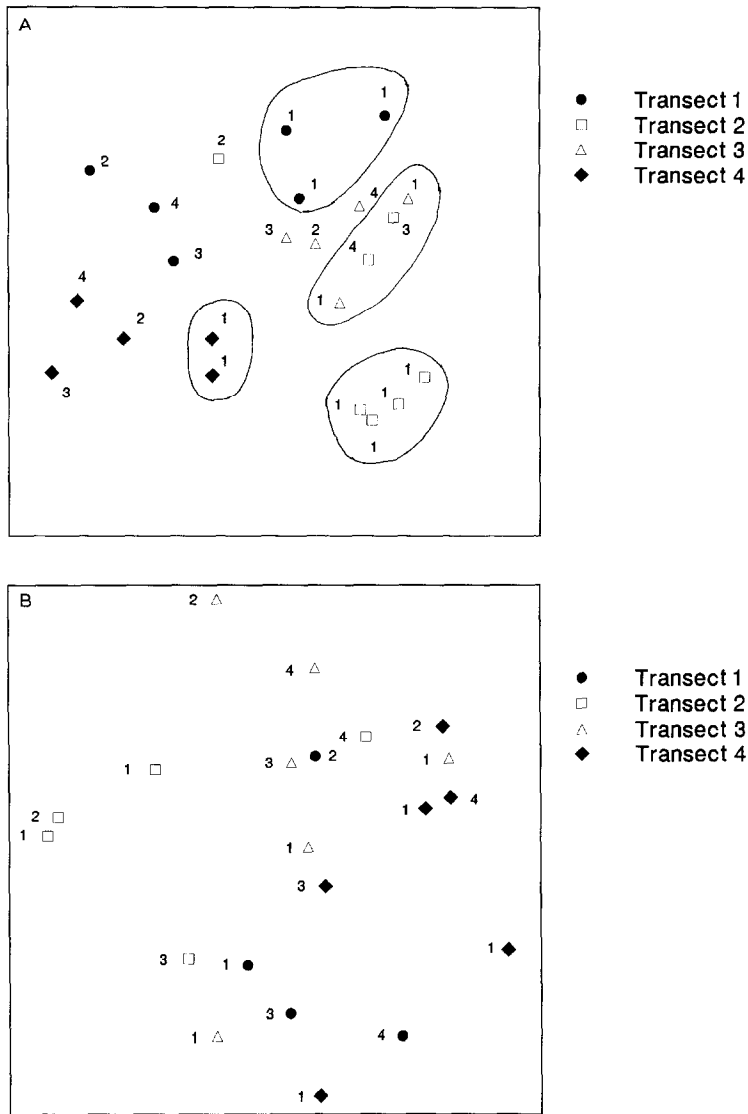


Fig. 4. Multi-dimensional scaling ordinations of a matrix of Bray-Curtis similarity indices based on square root-transformed species cover data from site 1 (A) and site 2 (B). Perimeters have been drawn around the pre-hurricane replicates of the four transects at site 1 which cluster independently from the post-hurricane samples of the same transects. Stress coefficients for the two plots were (A) = 0.15 and (B) = 0.12.

at the backreef site a decrease in coral cover of only approximately 2% was apparent. The north versus south reef differences reflect the southeasterly approach of the hurricane (Case & Mayfield, 1990). However, the depth to which extensive damage occurred was shallow compared with previous reports of severe storm impact (Highsmith

et al., 1980; Woodley et al., 1981; Laboute, 1985). Although Hurricane Hugo made direct landfall on St. Croix and the period of exposure to hurricane force winds was prolonged (12 h), it appears that average wave height was lower than expected (Hubbard et al., 1991), which may explain the relatively shallow zone of destruction.

All transects were separated by less than 1 m depth of water and adjacent parallel transects were only a few meters apart. Even so, one of the four transects at the south reef site (site 1) showed no significant hurricane damage. Similar patchiness has recently been reported for Hurricane Hugo damage in St. John (Edmunds & Witman, 1991; Rogers et al., 1992). At this scale, it seems unlikely that bathymetric or topographic variation in reef structure would have influenced wave action sufficiently to cause the differences observed (Kjerfve et al., 1986). Woodley et al. (1981) have shown that coral species may have different susceptibilities to storm damage. However, populations of the relatively fragile finger coral *Porites porites* survived intact on the single undamaged transect, a species which was severely impacted on the other transects. The patchiness might alternatively be explained by the damage being caused mainly by impact from loose debris rather than direct wave-action effects. The one transect which escaped serious damage could have done so either by chance, or due to a lack of suitable debris-forming substrate in the immediate surrounding area. Smaller-scale variability or patchiness in hurricane impact may have important consequences for post-impact recovery of community structure. Pockets of relatively undisturbed benthos may provide "seed populations" for recolonization of the surrounding area, either through fragment dispersal and propagation (Loya, 1976; Highsmith et al., 1980; Pearson, 1981) or via local dispersal and settlement of planula larvae (Gerrodette, 1981).

This study reinforces the need to work at several spatial scales when monitoring reef coral community structure, employing various techniques as suggested by Rogers (1988). Using haphazardly placed 1 m² quadrats, we were able to show quantitatively the between-zone differences discussed above (Gladfelter et al., 1991). The present analysis shows that monitoring fixed transects using the chain transect technique allows a more precise measurement of community structure to be made. Thus we were able to detect small changes in community structure which we were unable to do using haphazardly placed quadrats in the same area, analysed by univariate statistical measures. We have shown how even very small reductions in projected surface area of live tissues may represent a significant degree of mortality, particularly in long-lived, resilient corals such as *Diploria strigosa* (Bythell et al., in press). In addition, the use of fixed position transects enabled us to define the smaller-scale variability in hurricane impact on the scale of a few metres. Although the chain transect technique may offer greater precision, the sampling design was necessarily restricted because of the lengthy time requirements (up to 3 h per 20 m transect). Complementary data were therefore collected using both haphazard quadrats (low precision, high efficiency) and fixed chain transects (high precision, low efficiency).

Results of only two years' post-hurricane monitoring must be considered preliminary in long-lived coral populations (Rylandsdam, 1983; Babcock, 1988; Hughes & Jack-

son, 1980). However, the hurricane did not lead to an immediate increase in coral species diversity through differential mortality of the dominant species, as has been described elsewhere following hurricane impact (Porter et al., 1981; Woodley et al., 1981; Mah & Stearn, 1986; Rogers & Zullo, 1987). Populations of the dominant corals were reduced by the hurricane, particularly *P. porites* at site 1 although this was not significant due to spatial variability, but these species also recovered relatively rapidly during the ensuing months. Conversely, rarer species were also affected, and appear not to have recovered as rapidly and overall species richness was reduced. These results are very similar to those described previously for the impact of hurricanes David, Frederick and Allen on St. Croix (Rogers et al., 1983), even though they passed farther away than Hurricane Hugo with less severe impact.

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